

REAL ESTATE PRICE PREDICTION

SIC_AI801

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Executive Summary

This project develops machine-learning models to estimate Egyptian real-estate prices for two separate markets: property Sale and monthly Rent. The notebook starts from 39,713 listings with 53 variables, applies data-quality cleaning and leakage-safe preprocessing, engineers property and geographic features, and evaluates several regression models. Random Forest with enhanced property and location features is selected as the final approach for both markets.

1. Project Summary & Problem Statement

1.1 Project Overview

The business problem is to build a reliable automated reference for real-estate pricing. Given structured information about a property—such as area, bedrooms, bathrooms, property type, location, furnishing, amenities, listing status, coordinates, and listing date—the model predicts the property's price in EGP.

Because Sale prices and monthly Rent prices represent different markets with different price behavior, the notebook models them separately rather than forcing a single model to learn both markets.

Primary research/business question:

- Can structured property and location information be used to predict realistic Sale prices and monthly Rent prices in EGP?
- Which regression approach provides the strongest predictive performance while avoiding target leakage?
- Which property, location, and engineered features contribute most to the final predictions?

1.2 Dataset Description

Item	Details
Source	Kaggle — waddahali/real-estate-listings
Main file	propertyfinder.csv
Raw size	39,713 rows × 53 columns
Target	price_egp
Markets	Sale (buy) and Rent
Post-quality-gate size	39,586 rows × 31 columns before modeling feature engineering
Sale listings	19,802
Rent listings	19,783

Important raw variables include:

- Property information: property_type, area_value, bedrooms, bathrooms, furnished.
- Location: city, town, district, subdistrict, latitude (lat), longitude (lon).
- Listing information: completion_status, listing_level, is_premium, is_new_construction, is_direct_from_dev, is_exclusive.
- Presentation/amenity information: images_count, has_view_360, amenities.
- Time information: listed_date.
- The raw dataset also contains IDs, URLs, contact information, descriptions, titles, and other fields that are not used as predictive inputs.

2. Data Cleaning & Pre-processing

2.1 Data Cleaning Methodology

The notebook follows a staged cleaning process designed to preserve valid real-estate observations while removing unusable or clearly inconsistent records.

1. Duplicate checking: exact duplicated rows were checked and the result was 0. drop_duplicates() was still applied defensively.

- Reference audit: 126 rows were associated with duplicated non-null reference values. These were audited but not automatically removed because reference is not guaranteed to be a unique listing key.
- Completely missing/unsuitable fields were removed: video_url, rera, and agent_languages.
- Categorical missing values were explicitly represented as Unknown for fields such as completion_status, district, subdistrict, furnished, payment_method, title, location_full, reference, and amenities.
- Additional categorical fields were also protected with Unknown values: property_type, offering_type, price_period, city, town, and listing_level.
- price_egg rows with missing or non-positive values were removed.
- listed_date was converted to datetime so chronological modeling and time-based feature extraction could be performed.
- Bedrooms were normalized: studio → 0 and 7+ → 7, then converted to numeric.
- Bathrooms were normalized: none → 0 and 7+ → 7, then converted to numeric.
- A rental-market quality gate flagged 127 rental rows as inconsistent/extreme. The flags were based on sale-related title keywords and/or monthly rent above 2,000,000 EGP. These rows were removed before modeling.

After the quality gate, the modeling dataframe contained 39,586 rows and 31 columns. The remaining missing values were mainly 34 bedroom values and 29 bathroom values; these were handled later inside the modeling pipeline using training-data medians rather than filling them from the full dataset.

2.2 Outlier Detection and Treatment

The notebook used the IQR method to investigate price outliers. It identified 3,325 observations above the upper fence for price_egg. The extreme values were then inspected using the largest/smallest prices together with property type, area, and city.

The investigation showed that many high-price observations represented legitimate high-value properties such as villas, land, whole buildings, and palaces. Therefore, the project did not apply blanket statistical outlier deletion to Sale prices.

For Rent, the notebook additionally calculated an IQR fence (Q1 = 40,000 EGP, Q3 = 100,000 EGP, upper fence = 190,000 EGP) and clipped rent prices to that upper fence before the chronological split. This is distinct from the earlier rental quality gate: the quality gate removes clearly inconsistent records, while the clipping step limits the influence of extreme rental prices.

2.3 Feature Engineering

The notebook creates 20 core model features and an enhanced set of 30 features. The engineered variables are target-independent.

Engineered Feature	Purpose
amenities_count	Converts the amenities text into a numeric count of distinct amenities.
total_rooms	Combines bedrooms and bathrooms into a simple overall room-count signal.
listed_year	Captures broad time/market effects.
listed_month	Captures monthly/seasonal listing effects.
listed_dayofweek	Adds a small weekly time signal.
area_per_bedroom	Represents property size relative to bedroom count.
bathrooms_per_bedroom	Represents the relationship between bathrooms and bedrooms.
log_area	Represents the nonlinear effect of property size and reduces the dominance of very large areas.
lat_squared / lon_squared	Allows nonlinear geographic effects.
lat_lon_interaction	Captures an interaction between latitude and longitude.
location_grid	Creates a coarse local geographic grouping from rounded coordinates.
town / subdistrict	Adds finer location information beyond city/district.

The notebook explicitly verifies that the enhanced features do not contain forbidden/leakage fields. The leakage checks passed.

2.4 Data Transformation and Preprocessing

- Numeric missing values are imputed with SimpleImputer(strategy='median'). The imputer is fitted on training data only and then applied to validation/test data.
- Categorical missing values are imputed using the most frequent category.
- Categorical variables are converted using OneHotEncoder(handle_unknown='ignore', min_frequency=10). This prevents unseen validation/test categories from causing errors and limits extremely rare categories.

- The target price is transformed using $\log_{10}(\text{price_egp})$ during model fitting. Predictions are converted back to EGP using expm1 . This is appropriate for the strongly right-skewed price distributions.
- No feature scaling/normalization is applied to the Random Forest pipeline because tree-based models do not require distance-based scaling.
- The final workflow is implemented through preprocessing pipelines so transformations are learned from the appropriate training data.

2.5 Leakage Prevention

Target leakage was explicitly controlled. `price_per_sqm` is excluded because it is derived from the target price and would leak information about price into the model. Other excluded fields include `price_egp` itself, `category`, `offering_type`, `price_period`, `listing_type`, `scraped_at`, `listing_id`, `internal_id`, `detail_url`, `reference`, `description`, and `title`.

3. Exploratory Data Analysis (EDA)

3.1 Market Size and Typical Property Profile

Measure	Sale	Rent
Listings	19,802	19,783
Median area	166	186
Median bedrooms	3	3
Median bathrooms	3	3

The two markets are almost balanced in size. Typical properties have three bedrooms and three bathrooms. Rental listings have a slightly larger median area than Sale listings.

3.2 Descriptive Statistics and Distribution

Variable	Market	Mean	Median	Std	Max	Skewness
<code>price_egp</code>	Sale	12,550,000	9,100,000	$\approx 10.0\text{M}$	34,580,000	1.06
<code>price_egp</code>	Rent	83,945.54	65,000	$\approx 92.0\text{K}$	1,990,000	6.58
<code>area_value</code>	Sale	217.24	166	1096.69	96,600	74.53
<code>area_value</code>	Rent	219.48	186	154.69	3,500	5.32
<code>bedrooms</code>	Sale	3.05	3	1.21	7	0.39
<code>bedrooms</code>	Rent	2.85	3	1.10	7	0.29
<code>bathrooms</code>	Sale	2.99	3	1.32	7	0.72
<code>bathrooms</code>	Rent	2.81	3	1.17	7	0.80

The distributions are strongly right-skewed, especially Rent price and Sale area. This supports the notebook's use of log-transformed targets and explains why median values are useful alongside means.

3.3 Property-Type and Geographic Patterns

- Apartments represent about 53.07% of Sale listings and 73.90% of Rent listings.
- Villas represent about 18.04% of Sale and 10.39% of Rent listings.
- Chalets represent about 8.02% of Sale listings but only 0.36% of Rent listings.
- Townhouses represent about 7.64% of Sale and 3.36% of Rent listings.
- Cairo accounts for about 49.78% of Sale listings and 80.47% of Rent listings; Giza accounts for about 29.94% and 18.08%, respectively.
- The notebook's location analysis shows meaningful geographic price differences. North Coast has a high observed Sale median, while Cairo is a major contributor because of its large listing volume.
- Observed median Sale prices are particularly high for property categories such as Land and Palace, followed by Whole Building and Villa.
- For Rent, the observed median is particularly high for Chalet, followed by Villa and Twin House.

- Furnished rental properties show higher observed median rents than non-furnished properties in the notebook's analysis.

3.4 Correlations and Relationships

The notebook examines numeric relationships with price separately for Sale and Rent. For Sale, bedrooms and bathrooms show strong simple relationships with price, while area is also important. For Rent, area_value is the strongest simple numeric relationship with monthly rent. These correlations are useful for EDA, but they are not treated as the final measure of predictive importance because the Random Forest can model nonlinear effects and categorical/location interactions.

4. Machine Learning Model & Evaluation

4.1 Problem Classification

This is a supervised regression problem. The target variable is continuous price_egp. Two separate regression models are built: one for Sale prices and one for monthly Rent prices.

4.2 Train / Validation / Test Strategy

Because real-estate prediction should reflect future prediction rather than random mixing of old and new listings, the notebook sorts observations by listed_date and uses a chronological split:

Split	Share	Purpose
Training	70% oldest listings	Fit preprocessing and model parameters.
Validation	Next 15%	Compare models, tune Random Forest, and test feature variants.
Test	Newest 15%	Final one-time evaluation after model selection.

Sale: 13,861 train, 2,970 validation, 2,971 test. Rent: 13,848 train, 2,967 validation, 2,968 test. The test set is kept untouched during feature engineering and hyperparameter selection.

4.3 Models Experimented With

Model	Role / Reason for Experimenting
Linear Regression	Simple baseline; tests whether a mostly linear relationship is sufficient.
Random Forest Regressor	Captures nonlinear relationships and interactions among structured property/location features.
Gradient Boosting Regressor	Alternative sequential tree-based nonlinear model.
CatBoost Regressor	Categorical-friendly boosting approach and useful benchmark for mixed categorical/numeric data.

4.4 Benchmark Results

Market	Model	MAE (EGP)	RMSE (EGP)	R ²
Rent	Random Forest	12,735.64	20,135.19	0.8309
Rent	CatBoost	13,989.03	21,283.33	0.8110
Rent	Gradient Boosting	18,524.35	27,265.91	0.6899
Rent	Linear Regression	20,565.64	32,058.82	0.5712
Sale	Random Forest	2,433,652	3,930,593	0.8362
Sale	CatBoost	2,673,494	4,090,217	0.8226
Sale	Gradient Boosting	3,390,844	4,905,506	0.7449
Sale	Linear Regression	Very unstable	Very unstable	Extremely negative

The benchmark indicates that tree-based nonlinear models are substantially better suited to the structured real-estate problem than Linear Regression. Random Forest is the strongest benchmark among the tested models for both markets.

4.5 Targeted Hyperparameter Tuning

Random Forest was tuned using a targeted set of sensible configurations. Validation R^2 was the primary selection criterion, while MAE and RMSE were retained as secondary error measures.

Market	n_estimators	max_depth	min_samples_leaf	max_features	Validation R^2
Sale	200	None	2	0.8	0.8465
Rent	250	20	1	0.8	0.8353

4.6 Feature Engineering Experiment

Market	Feature Set	MAE	RMSE	R^2
Sale	Core	2,403,373	3,894,627	0.8392
Sale	Enhanced property + location	2,344,845	3,804,820	0.8465
Rent	Core	12,341.34	19,728.75	0.8376
Rent	Enhanced property + location	12,403.10	19,869.41	0.8353

Enhanced features improved Sale validation R^2 from 0.8392 to 0.8465. For Rent, the enhanced set did not improve the validation score (0.8376 core vs 0.8353 enhanced). Nevertheless, the notebook's final workflow reports the enhanced feature configuration for both markets.

4.7 Final Test Evaluation

Market	Final Model	MAE	RMSE	R^2
Sale	Random Forest + enhanced features	2,496,186.60 EGP	3,975,698.09 EGP	0.83
Rent	Random Forest + enhanced features	11,239.98 EGP	18,444.98 EGP	0.84

R^2 measures the proportion of price variance explained by the model. MAE represents the average absolute prediction error in EGP. RMSE penalizes large errors more strongly than MAE.

4.8 Business Baseline Comparison

Market	Approach	Reference	MAE	RMSE	R^2
Sale	Median baseline	9,000,000 EGP	7,473,195.03	10,470,093.15	-0.19
Sale	Final Random Forest	Model	2,496,186.60	3,975,698.09	0.83
Rent	Median baseline	65,000 EGP	34,805.75	46,566.50	-0.02
Rent	Final Random Forest	Model	11,239.98	18,444.98	0.84

The final models clearly outperform a naive median-price reference. For Rent, the notebook reports a 67.71% improvement in MAE relative to the median baseline.

4.9 Error Analysis

The notebook evaluates prediction errors across five price bands. Sale errors increase in absolute EGP terms as the price band increases, while Rent errors also rise toward the highest-price segment.

Market	Band	MAE	Within 20%
Sale	Lowest	1,137,107	48.85%
Sale	Low	1,292,898	65.64%
Sale	Middle	1,992,243	59.90%
Sale	High	3,436,708	57.91%
Sale	Highest	4,631,731	68.01%
Rent	Lowest	6,183.61	60.73%

Rent	Low	7,408.92	73.28%
Rent	Middle	8,959.73	77.54%
Rent	High	12,183.94	76.99%
Rent	Highest	21,968.82	73.23%

4.10 Feature Importance

The final Random Forest feature-importance analysis highlights the following variables.

Market	Feature	Importance
Sale	log_area	0.2956
Sale	area_value	0.2874
Sale	lat_lon_interaction	0.0416
Sale	property_type_Apartment	0.0331
Sale	amenities_count	0.0277
Sale	area_per_bedroom	0.0229
Rent	area_value	0.2287
Rent	log_area	0.2027
Rent	furnished_NO	0.0944
Rent	lat_lon_interaction	0.0334
Rent	total_rooms	0.0281
Rent	area_per_bedroom	0.0276
Rent	bathrooms_per_bedroom	0.0247

The importance results reinforce the role of property size and location. Sale predictions are especially influenced by log-transformed area and raw area, while Rent predictions also show a notable furnishing effect.

5. Conclusion, Recommendations & Future Work

5.1 Main Findings

- The dataset contains nearly equal numbers of Sale and Rent listings, allowing both markets to be modeled separately.
- Real-estate price distributions are strongly right-skewed, particularly monthly Rent and Sale area.
- Property size, layout, property type, furnishing, amenities, and geographic information are important predictive signals.
- Random Forest substantially outperformed Linear Regression and Gradient Boosting in the benchmark and was also stronger than CatBoost in the executed benchmark results.
- The final Random Forest achieved $R^2 = 0.83$ for Sale and $R^2 = 0.84$ for Rent on the untouched chronological Test set.
- The models significantly outperform simple median-price baselines.
- Rental forward-validation results remained stable, with R^2 from 0.82 to 0.85.
- The notebook explicitly checks for target leakage and excludes price_per_sqm because it is target-derived.

5.2 Business Recommendations

- Use the final models as automated reference-pricing tools for agents, property platforms, and internal market analysis.
- Use model estimates as a decision-support reference rather than an absolute valuation, especially for unusual luxury properties and very high-value Sale listings.
- Collect and maintain accurate area, property type, furnishing, room counts, and detailed location fields because these variables strongly influence predictions.
- Monitor the model over time and retrain periodically as the real-estate market changes.
- For high-value or unusual listings, consider an additional review workflow because absolute errors naturally become larger at higher price levels.

- Continue validating Rent performance through forward time windows because rental prices can change with market conditions.

5.3 Limitations

- The dataset is based on online property listings, so listing prices may not equal final transaction prices.
- The model depends on the quality and completeness of listing attributes; missing or inaccurate property details can affect predictions.
- Very high-value properties and unusual property types are harder to predict in absolute EGP terms.
- The final evaluation is based on a particular chronological test period, so performance may change under future market conditions.
- The notebook's rental preprocessing includes clipping prices at the calculated IQR upper fence of 190,000 EGP; this improves robustness to extreme values but also means the final Rent model is optimized for the clipped target distribution.
- The feature-engineering experiment did not improve Rent validation R^2 , so the value of every enhanced feature is not universal across markets.

5.4 Future Work

11. Evaluate additional categorical-friendly models and stronger boosting approaches under the same chronological protocol.
12. Perform broader hyperparameter optimization while preserving the untouched test set.
13. Add richer temporal features and monitor concept drift as new listings arrive.
14. Investigate robust treatment of extreme Sale properties rather than removing legitimate high-value observations.
15. Compare predictions against actual transaction/lease outcomes if such ground-truth data become available.
16. Add uncertainty estimates or prediction intervals so stakeholders can see the expected range around a point estimate.
17. Build a deployment dashboard where users enter property characteristics and receive Sale and Rent reference estimates.
18. Monitor production errors by market, location, property type, and price segment to identify where retraining or additional data are needed.

Appendix A — Final Modeling Feature Sets

Core features (20): area_value, bedrooms, bathrooms, total_rooms, property_type, furnished, city, district, lat, lon, amenities_count, completion_status, listing_level, is_premium, is_new_construction, is_direct_from_dev, is_exclusive, images_count, listed_year, listed_month

Enhanced features (30) add: area_per_bedroom, bathrooms_per_bedroom, log_area, lat_squared, lon_squared, lat_lon_interaction, location_grid, town, subdistrict, listed_dayofweek

Appendix B – Final reference estimator

The notebook includes a working reference-price estimator. For an example property with area 170, 3 bedrooms, 2 bathrooms, Apartment type, furnished = YES, Cairo / The 5th Settlement, the model produced approximately:

Output	Estimated Value
Sale price	6,857,432 EGP
Monthly Rent	56,077 EGP